UNITED STATES · NUCLEAR CONCEPTUAL PLANNING GUIDE

A new nuclear plant design and implementation guide that can meet NRC and financing requirements to remain profitable

Evaluation of known designs, and a proposed U.S. Forge-350 plant with an implementation path.

U.S. Forge-350

Fluoride-salt-cooled, TRISO-fueled coal-site repower plant with decoupled nitrate-salt storage

Author: Keith J Maxwell, October 2026 · Conceptual design for readers, utilities, and investors. Not an NRC submittal, safety analysis, or offer to sell securities.

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Readers seeking technical information on reactor or fuel-cycle technologies should consult the relevant licensing authorities and reactor vendors and comply with all applicable export control regulations.

1. How to use this guide

This guide does two things. It judges the nuclear plant types that are actually known — operating, licensed, or far enough along to have a safety case — against the four constraints that decide whether a plant gets built in the United States: thermal and fuel efficiency, the safety case a regulator can review, the licensing path that exists in 2026, and the economics of first-of-a-kind and later units. It then outlines one plant concept, Forge-350, at the program level, and the sequence of business and licensing decisions required to take that plant from a site screen to commercial operation.

The judgment is for a grid-scale plant whose owner wants a construction permit this decade and commercial operation in the mid-2030s. It is not a ranking of scientific elegance. A liquid-fuel thorium reactor can be the best physics and still be the wrong next plant, because the United States has no operating license precedent, no qualified salt-facing code case in commercial use, and no fuel-cycle plant ready to support it.

Status of this document. Forge-350 is a synthesis for planning. It is not a preliminary safety analysis report, not an NRC endorsement, not a reactor design, and not a cost estimate an investor can take to a credit committee without an owner's engineer. Figures are order-of-magnitude business and schedule targets anchored to public 2024–2026 project data. Tax credits, fee rules, and hearing practice change; confirm them with counsel before filing.

2. What actually decides a U.S. project

The operating U.S. fleet is the evidence that the technology works. In 2025 nuclear posted a 91 percent capacity factor, ahead of every other major source, from a fleet of 94 reactors and about 97 GW. Roughly two-thirds are pressurized water reactors and the rest boiling water reactors. That fleet is also the evidence that a new plant can still fail as a business. Plant Vogtle Units 3 and 4, the only U.S. reactors completed from a cold start in a generation, came online in 2023–2024 about seven years late. Reported all-in cost is on the order of $31–37 billion for 2,234 MW, about $14,000–16,000 per kilowatt, against an early certification near $14 billion. The same AP1000 design in China was delivered for on the order of $10 billion for four units' worth of earlier pairs, on roughly half the U.S. schedule. The design was not the whole difference. Supply chain, design completion, interest during construction, and a single over-matched EPC contract were.

Four levers follow from that record, and they structure the rest of this guide.

- **Efficiency** is not only thermal percentage. A plant at 33 percent that runs at 93 percent capacity factor and uses a fuel cycle that already exists beats a 50 percent plant that cannot be fueled. Count electrical efficiency, fuel utilization, capacity factor, and usable heat.
- **Safety** in the U.S. sense is a reviewable argument: defense in depth, a mechanistic source term, passive decay-heat removal, and a design basis the staff has methods for. Walk-away language does not replace a transient analysis.
- **Regulation** as of October 2026 still runs through 10 CFR Part 50 (construction permit, then operating license) or Part 52 (design certification and combined license). Part 53, the risk-informed advanced-reactor framework enabled by the Nuclear Energy Innovation and Modernization Act and pushed by the ADVANCE Act, is not yet the production path. The proven 2025–2026 path for a first unit is a Part 50 construction permit.
- **Economics** are dominated by overnight capital and interest during construction. Fuel is a few dollars per megawatt-hour on a light-water plant. A five-year slip at 8 percent money can erase the entire fuel advantage. Serial build, a finished design, a reused coal site, and a contract structure that does not bet the project on one EPC are the available levers.

3. The 2026 U.S. baseline

The licensing picture is no longer theoretical. In March 2026 the NRC issued a construction permit for TerraPower's Natrium reactor at Kemmerer, Wyoming — the first commercial non-light-water construction permit in more than forty years. Natrium is a sodium-cooled fast reactor rated at 345 MWe, with nitrate-salt storage intended to raise output to 500 MWe for more than five hours. DOE cost share is up to about $2 billion of a roughly $4 billion project. Nuclear construction began in 2026; the sponsor has targeted completion near 2030 and commercial operation near 2031.

In September 2026 the EPC contractor, Bechtel, and TerraPower parted ways and the EPC scope was put back out to bid, with a Hyundai Engineering & Construction framework discussed for later units. That split is a live lesson, not a footnote.

On 29 September 2026 the NRC issued a construction permit for the Tennessee Valley Authority's GE Vernova Hitachi BWRX-300 at Clinch River, after a review of about fourteen months. The BWRX-300 is a 300 MWe boiling-water small modular reactor derived from the ESBWR certification basis, using standard low-enriched uranium. Public discussion of TVA's first-unit cost has cited figures near $5.4 billion before interest, on the order of $18,000 per kilowatt, with later units hoped to fall toward $3.7 billion. A headline levelized cost near $60 per megawatt-hour only works if capital is far below those first-unit figures. The first BWRX-300 unit is already under construction at Darlington in Ontario.

Kairos Power has safety-related construction underway on Hermes, a small fluoride-salt-cooled test reactor at Oak Ridge using TRISO pebble fuel and a molten-salt coolant at low pressure. Hermes 2, the first power-producing unit in that line, broke ground in 2026 under a power agreement involving Google and TVA. X-energy's Xe-100 high-temperature gas reactor is in construction-permit review for Dow's Seadrift, Texas site, and its TRISO-X subsidiary holds an NRC Part 70 license for HALEU fuel fabrication — the only such commercial fabrication foothold. NuScale remains the only SMR with a design certification and a later standard design approval (the 77 MWe US460, May 2025). Its first U.S. project, the Carbon Free Power Project, was cancelled in 2023 after the estimate reached about $20,000 per kilowatt.

Fuel is the constraint under all of the non-light-water projects. High-assay low-enriched uranium (HALEU), the regulatory category above conventional reactor fuel and below highly enriched uranium, is required by Natrium, Kairos, and X-energy. Centrus is licensed to produce limited quantities of HALEU at Piketon; that output is pilot scale relative to commercial demand, which is measured in tonnes, not hundreds of kilograms. The ADVANCE Act tells the NRC not to unnecessarily limit civilian use, cuts some fees, and pushes faster reviews for brownfield and retiring-coal sites. House permitting bills advanced in September 2026. None of that creates enrichment capacity. A plant that needs HALEU must contract for it before it asks for a permit.

4. Evaluation of known designs

Scores below are 1 (poor fit) to 5 (strong fit) for a U.S. grid plant aimed at a construction permit before 2030. They are planning judgments, not physics grades. "Licensable now" means a staff review can start from existing rules, precedent, and fuel, not that a permit is assured.

Family	Efficiency	Safety maturity	Licensable now	U.S. fuel	FOAK cost	NOAK path
Gen III+ PWR (AP1000)	2 — ~33%	5 — operating	5 — certified	5 — LEU	1 — Vogtle	3 — if design frozen
BWR SMR (BWRX-300)	2 — ~33%	4 — ESBWR basis	5 — CP issued	5 — LEU	2 — still FOAK	4 — Canada lead
CANDU / PHWR	2 — ~31%	4 — decades	2 — no U.S. chain	3 — nat. U, D2O	2	3 — wrong supply base
Sodium fast + storage	3 — ~40%	3 — EBR-II, not recent U.S.	4 — CP issued	2 — HALEU metal	2 — ~$12k/kW target	4 — if fuel arrives
Salt-cooled TRISO (FHR)	4 — 42% class	3 — MSRE + TRISO	3 — test CP only	2 — HALEU TRISO	2	4 — compact island
HTGR (Xe-100 class)	4 — heat + power	4 — German/U.S. tests	3 — CP in review	2 — HALEU TRISO	2	3 — lower power density
Liquid-fuel MSR	5 — temp. and burnup	2 — no commercial case	1	1	1	4 — if ever licensed
Lead- or lead-bismuth	4	2 — materials	1	2	1	3
Microreactor	2	3	3 — DOE/NRC split	2	2 — per kW high	3 — niche only

Figure. Fit for a U.S. grid plant seeking a permit this decade. A 5 is not "best reactor." It is "least likely to be the reason the project stops."

4.1 Light-water reactors

The pressurized-water reactor is the default U.S. machine: a high-pressure water-cooled design with net efficiency of about one-third and refueling outages every year and a half to two years. The AP1000 added passive safety systems and a certified design. Safety maturity is the best in the table. Economics are the worst recent U.S. data point, and the cause is knowable: an incomplete design at first concrete, a hollow supply chain, a consortium that failed (Westinghouse's 2017 bankruptcy), and interest on a decade of construction. A second U.S. AP1000 pair, on a finished design with the Vogtle workforce still warm, is a different project from Vogtle. It is not a different reactor. It does not raise efficiency, it does not shrink the seismic civil works, and it still asks an owner for several billion dollars before the first kilowatt-hour.

The BWRX-300 is the light-water design that best matches a 2026 U.S. start. It simplifies older boiling-water plants and inherits the passive safety approach of the certified ESBWR. Fuel is standard LEU, so it does not wait on HALEU. TVA's construction permit shows the NRC can review it on a utility schedule. The efficiency penalty remains: about one-third of the fission heat becomes electricity, and the plant cannot sell high-temperature process heat. First-unit capital, if the public

figures hold, does not by itself beat a gas plant on energy-only price. It can still be the right plant where an owner needs firm megawatts before HALEU exists in tonnes.

NuScale's integral PWR module is the only design-certified SMR. Certification did not save the first project. Factory fabrication still has to be cheap at a volume no U.S. shop has produced, and a 77 MWe module spreads owner's costs over too few megawatts unless many modules share a building. Integral PWRs remain a credible product. They are not the cost escape that was advertised in 2020.

4.2 Heavy water

CANDU and other pressurized heavy-water reactors burn natural uranium, refuel on line, and have a long operating record in Canada, Korea, China, and elsewhere. Their fuel economy is excellent. Electrical efficiency is not — about 30 percent — and the heavy-water inventory is a capital and tritium burden. The United States has no commercial heavy-water supply chain and no licensed CANDU. Online refueling also raises a safeguards conversation the U.S. program does not need for a first new plant. CANDU is a good reactor in the wrong country.

4.3 Gas-cooled and high-temperature reactors

Britain's AGRs showed that carbon-dioxide cooling at high outlet temperatures can deliver net efficiencies around 40 percent. They also showed graphite and steel aging under decades of operation. The modern heir is the high-temperature gas reactor: helium coolant and TRISO fuel in pebbles or prismatic blocks. TRISO particles — small fuel kernels sealed inside multiple ceramic and carbon coatings — are designed to retain fission products at temperatures well above normal operation. That is the strongest solid-fuel source term available. X-energy's Xe-100, aimed at about 80 MWe per module and at Dow process steam as well as power, is the U.S. commercial carrier of this idea, with a fabrication license already in hand.

The limits are power density and the pressurized helium boundary. Gas carries less heat per unit volume than liquid salt or sodium, so the core and the building are large for the megawatts. A pressurized gas system reintroduces a depressurization accident that liquid-salt designs are built to avoid. For a refinery or chemical host that wants high-temperature process steam, an HTGR is often the better fit. For a utility replacing a coal unit on a megawatt basis, it is a slower way to accumulate capacity.

4.4 Sodium-cooled fast reactors

Natrium is the design to beat on project structure, not on coolant. A fast-spectrum design opens a later path to higher fuel utilization. Its higher operating temperature supports a respectable steam cycle, about 40 percent. The storage idea — reactor at steady power, nitrate-salt tanks selling the peak — is the right answer to a grid that already has variable wind and solar and that pays for capacity. Low primary pressure removes the classic loss-of-coolant break.

Sodium's problems are equally specific. It burns in air and reacts violently with water, so the plant needs extra isolation systems and a public story the U.S. has not had to tell since the Fast Flux Test Facility and EBR-II shut down. Fast-reactor safety questions must be resolved in the vendor's design and the NRC's review, not asserted away. Commercial-scale fabrication of its HALEU fuel does not yet exist. Kemmerer is the demonstration that answers these questions. Forge-350 copies Natrium's commercial architecture and does not copy its coolant, because a first fleet does not need a sodium-water reaction on the risk register if another low-pressure coolant can do the job.

4.5 Salt-cooled solid fuel, and liquid-fuel molten salt

These are different reactors that share a word. A fluoride-salt-cooled high-temperature reactor keeps fuel inside TRISO pebbles and uses Flibe — a lithium-beryllium fluoride salt — only as coolant. Hermes is licensed to operate at low pressure, and the salt's boiling point sits far above its operating temperature, so the margin to boiling is large. The salt also retains many fission products that escape a particle, which is why Kairos treats TRISO plus Flibe as functional containment. Net electrical efficiency in the 40 percent class is a fair planning target. Kairos's public licensing documents describe passive, natural-circulation decay-heat removal.

A liquid-fuel molten salt reactor dissolves the fuel in the salt. That gives the best combination of temperature, online chemistry, and eventual fuel use, including a thorium cycle. It also puts the source term in a form that can leak and requires additional safety-class systems. The Molten Salt Reactor Experiment at Oak Ridge ran in the 1960s; it was not a licensed power plant, and its materials questions were not closed. No liquid-fuel design has a U.S. construction permit.

Materials qualification under ASME Section III Division 5, safeguards on a circulating fuel salt, and a chemical plant inside the nuclear island are a 2040s program unless a test reactor changes the record. Thorium does not fix the 2030s fuel-supply problem: starting a thorium cycle still needs an existing fissile fuel supply.

4.6 Lead-cooled reactors and microreactors

Lead and lead-bismuth fast reactors offer a high boiling point and no chemical fire. They also offer corrosion, erosion, and — for lead-bismuth — polonium-210. Russian submarine experience is not a U.S. license basis. No U.S. commercial project is ahead of paper on this coolant. Microreactors (Oklo's line, Radiant's Kaleidos at about 1 MWe, defense transportable units) matter for mines, bases, and remote grids. They do not replace a retired 400–800 MW coal unit, and their cost per kilowatt is structurally high.

4.7 Fusion, as a source rather than a plant

Fusion is not in the implementation path of this guide. No device has yet run a sustained net-electric fuel cycle with a qualified blanket, tritium accountancy, and a licensed site for power export. The ADVANCE Act's separate fusion framework is useful so that fusion pilots are not forced through power-reactor rules written for fission. It does not put firm megawatts on a 2035 resource plan. A reader who needs energy this decade should not trade a fission permit for a fusion option.

5. Chinese technology, supply chain, and labor

China is the only country building nuclear plants in volume. That fact belongs in a U.S. implementation guide, and so do the rules that stop a direct transplant. The cost gap is real. The legal path for Chinese hardware, a Chinese designer of record, or a Chinese construction crew on a U.S. nuclear island is not.

5.1 What China actually fields

The export machine is the Hualong One (HPR1000), a large pressurized-water reactor of roughly 1,100 MWe-plus with active plus passive safety systems. CNNC has quoted a domestic construction cost of about 17,000 yuan per kilowatt — on the order of $2,400 per kilowatt at recent exchange rates. Treat that as a vendor figure for a domestic overnight cost, not a U.S. all-in price. Fuqing 5 and 6, Fangchenggang 3 and 4, and later units are operating or under construction; State Council approvals in 2024, 2025, and 2026 kept the line full. Pakistan is the only real export: Karachi units financed mostly with Chinese loans, plus a further Chashma unit. The U.K. Generic Design Assessment was completed in 2022, and Bradwell still was not built. Argentina's Atucha III contract stalled on financing and fuel. A passed safety review in a Western regulator did not produce a Western plant.

CAP1000 and CAP1400 are the domestic descendants of the Westinghouse AP1000 built at Sanmen and Haiyang. Those four units are the clean comparison with Vogtle: same reactor, on the order of $10 billion for the Chinese four-unit program against more than $30 billion for two U.S. units, and about five to six years to grid connection against about eleven. Westinghouse's own account of the later Chinese AP1000s is that they now go from start to finish in under five years because the design is finished and the supply chain is warm. That is a construction result, not evidence that the plant is a different machine.

Two advanced units matter as data, not as import candidates. Shidaowan's HTR-PM is an operating pebble-bed high-temperature reactor, the closest thing to commercial TRISO experience at power-plant scale. Linglong One (ACP100), a 125 MWe integral PWR, started construction in July 2021 as the first land-based commercial SMR and was still in pre-fuel commissioning in September 2026, with commercial operation slipping toward 2027. China's SMR has not escaped a first-unit slip. A sodium-cooled CFR-600 demonstration fast reactor is a military-adjacent fuel-cycle project, not a template for a U.S. utility.

5.2 Why the U.S. price is higher, and what of that is copyable

The Sanmen–Vogtle gap is the useful Chinese evidence. It does not come from a secret coolant. It comes from a State Council that orders reactors in batches, policy-bank finance, a design that is not redrawn after first concrete, module yards, a heavy-forge industry that is already qualified, and a labor rate and a regulatory cadence the United States does not have. Serial Hualong One construction is what a U.S. fleet order would be trying to imitate.

Chinese advantage	Survives contact with U.S. law?	What to copy instead
Batch approval of identical units	Yes, as a commercial act	Order 4–8 identical units before nuclear concrete; freeze the design
Finished design, under-five-year AP1000 builds	Yes	No first concrete on an open drawing set — the Vogtle failure mode
Domestic heavy forgings (CFHI, Erzhong, Shanghai Electric)	No, as nuclear-island safety class	Doosan, Japan Steel Works, or a new U.S. press under NQA-1
Policy-bank debt and state owner	Only the function	DOE Title 17, regulated rate base, or a federal offtake — not a Chinese loan
Lower construction wages	No	Prevailing wage is the price of the tax credit; productivity, not wage arbitrage
NNSA review cadence and limited intervention	No	Part 50 two-step permit, as demonstrated in 2026; hearings still exist
Hualong One vendor cost claim ~$2,400/kW	Not as a U.S. bid	Use it as a floor the U.S. will not hit, not as a pro forma

Figure. Which parts of the Chinese build model can be used on a U.S. site. The cost gap is mostly the rows that do not transfer.

5.3 The license barrier

A utilization-facility license cannot be held by an entity owned, controlled, or dominated by a foreign government unless a specific exception applies. In April 2026 the NRC amended 10 CFR 50.38 to implement ADVANCE Act section 301. The exception covers listed OECD countries and India, and it still requires a finding that issuance is not inimical to the common defense and security or public health and safety. China is not on that list. CNNC, CGN, and SPIC are state-owned. They cannot be the licensee, and a structure in which they control the safety case, the fuel, or the operating organization will fail the foreign ownership, control, or domination test. A negation action plan of the kind used for allied minority investors is not a path to a Chinese designer of record.

CGN has been on the Commerce Department Entity List since August 2019. In 2016 the Justice Department charged CGN and an intermediary in a conspiracy to develop special nuclear material in China without U.S. authorization; the public FBI account describes a long campaign to collect U.S. reactor technology through commercial joint ventures. CGN is also carried on the U.S. list of Chinese military companies. That history is why a “technology partnership” with a Chinese national champion is a security finding waiting to happen, not a procurement strategy. Equipment and services with a CGN nexus also collide with export-control and deemed-export rules if U.S. technology flows back.

5.4 Supply chain

Chinese mills can forge a reactor vessel. The question is whether a U.S. project can accept it. Safety-class components need an ASME nuclear stamp, an NQA-1 quality program the NRC can inspect, and a vendor that will open its shop to U.S. inspectors. A vessel built under a Chinese nuclear QA program does not arrive pre-accepted. Requalifying one is a multi-year detour that spends the schedule the import was supposed to save.

Tax law now prices the same choice. The 2025 reconciliation act extended prohibited-foreign-entity rules to the section 45Y and 48E credits. Specified foreign entities — including the government of China, Chinese military companies, and entities organized in China — cannot claim those credits. For facilities beginning construction after 2025, material assistance from a prohibited foreign entity above a falling threshold disallows the credit. IRS Notice 2026-15 sets the interim computation; for 2026 the qualified-facility threshold is on the order of 40 percent non-prohibited content, tightening after that. A Chinese nuclear steam supply system is not a rounding error in that ratio. An owner who wants the credit cannot fill the nuclear island from a covered-nation vendor. Fuel is the same problem: commercial U.S. cores are fed by allied enrichment, and a Chinese fuel contract would be a Part 110 and political non-starter.

The allied substitute already exists and was used at Vogtle. Doosan supplied vessels and steam generators; Japan Steel Works remains the reference heavy-forge shop. “We buy where we build,” in Westinghouse’s formulation, is the legal version of China’s domestic chain. A U.S. implementation plan should option Korean and Japanese long-lead forgings at Gate 1 and treat a new domestic press as a fleet investment, not as a unit-1 critical path.

5.5 Labor

Chinese nuclear construction productivity is real, and a Chinese crew on a U.S. site is not the way to buy it. Unescorted access inside a licensed plant is an NRC access-authorization decision under 10 CFR 73.56, with a background investigation. A workforce drawn from a covered nation, employed by a state-owned nuclear enterprise, will not clear that program at the scale of a civil megaproject. H-2B and similar visas are capped far below a nuclear payroll and are a poor fit for multi-year nuclear work. On a credit-qualified project, prevailing-wage and apprenticeship rules set the wage. After that rule, imported labor is not cheaper; it is a second compliance system.

What does transfer is method. China's crews are fast because they have built the last unit, because the design is frozen, and because modules arrive complete. A U.S. owner reproduces that by keeping the Vogtle craft pipeline employed, by writing a multi-unit order so the crew does not disband, and by paying for a module yard. Hiring CNNC's construction arm would also be a deemed-export and FOCD problem the moment those workers needed the safety drawings.

5.6 What an implementation may take from China

- **Take the batch.** Do not license a one-off. Name four to eight identical units in the first filing wave, the way the State Council approves Hualong pairs, so the factory and the crew have a second unit waiting.
- **Take the finished design.** The under-five-year Chinese AP1000 is the evidence that schedule follows drawing freeze. Forge-350 does not pour nuclear concrete on an open design.
- **Take the operating data, not the license.** HTR-PM operating experience is relevant to a TRISO safety case. It enters a U.S. application only after it is re-analyzed under a QA program the NRC can credit. It is not a Chinese design certification.
- **Do not take the vendor, the vessel, the fuel, or the crew.** Each collides with 50.38, the Entity List, prohibited-foreign-entity credit rules, or access authorization. Any one of those is enough to erase the wage and forge savings.

> **Planning rule.** If a proposal needs a Chinese state-owned designer, a Chinese safety-class vessel, Chinese fabricated fuel, or a Chinese nuclear construction force in order to hit its price, it is not a U.S. project. Rewrite it as a U.S. or allied design, allied forgings, and a domestic crew on a frozen multi-unit order — or stop.

6. What the evaluation selects

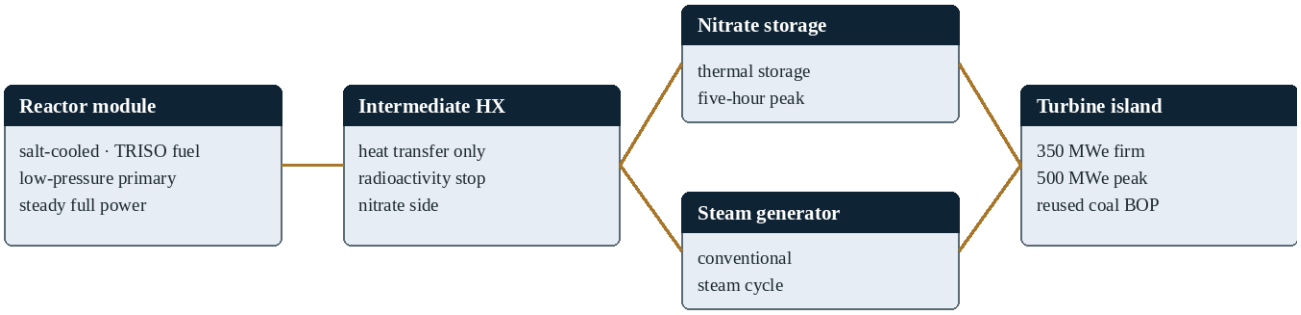
The plant that best joins the four constraints is not the plant with the highest score in any single column. It is a fluoride-salt-cooled, TRISO-fueled module, sited on a retiring coal plant, with Natrium's storage split and the BWRX-300's discipline about not inventing a fuel cycle.

- **From light water:** a finished safety case culture, a two-step Part 50 permit, and a refusal to make the first unit depend on a new enrichment category if the salt-fuel line slips. The guide therefore carries an explicit off-ramp to a BWRX-class island on the same site.
- **From Natrium:** coal-site reuse, DOE cost share as a FOAK tool, and thermal storage so the reactor does not load-follow. Also the negative lesson of September 2026: do not let one EPC own the only version of the schedule.
- **From Kairos:** a low-pressure salt coolant, TRISO functional containment, and a passive decay-heat approach, all already on the NRC docket. Hermes must produce the fuel and salt data before a commercial operating license; the commercial application should be written so that it can incorporate that data without a redesign.
- **From X-energy:** a domestic TRISO fabrication license is a strategic asset. Forge-350 should be able to buy fuel from more than one qualified shop, starting with the shop that already has a Part 70 license.
- **Rejected for the first plant:** liquid fuel, thorium as the main cycle, lead coolant, a full closed fuel cycle, and any design that needs a new pressure-vessel forging the United States cannot deliver before 2032.

7. Forge-350 plant description

Forge-350 is a planning concept for a single-module energy center at coal-unit scale: about 350 MWe of firm output, with nitrate-salt thermal storage that lets the plant sell 500 MWe for about five hours at peak while the reactor stays at steady power. Its architecture follows public advanced-reactor descriptions — a low-pressure, salt-cooled reactor with TRISO

fuel, an intermediate loop that separates the reactor from the storage and power cycle, and a conventional steam turbine island — on a coal site whose switchyard, rail, cooling makeup, and transmission interconnection already exist. Reactor design parameters are deliberately not specified here; they belong to a licensed vendor's design and an NRC application, not to a public planning guide.



Heat only crosses the intermediate exchanger. The power cycle and the storage tanks never see primary salt. Peak power is drawn from storage, so the reactor stays at steady full power.

Figure. Forge-350 heat path, conceptual. Storage and the steam cycle sit on the intermediate nitrate loop, not on the primary salt.

Attribute	Forge-350 planning position	Why this position
Reactor	Salt-cooled, TRISO-fueled advanced reactor, one module, from a vendor with an NRC docket	Avoids multi-module FOAK coordination; matches coal-unit scale
Coolant	Fluoride salt, per the selected vendor design	Low pressure; follows a licensing line the NRC has already reviewed
Fuel	TRISO pebble fuel from a U.S.-licensed fabricator	Robust fuel form; fabrication already licensed at one U.S. shop
Fuel supply	HALEU, contracted before the permit filing	Fuel availability, not physics, sets the schedule
Test-data basis	Within the envelope covered by the Hermes test program	So public test data transfers to the commercial application
Primary system	Low pressure	Removes high-pressure blowdown as the sizing accident
Intermediate	Nitrate solar salt	Chemical isolation plus the storage medium; CSP operating experience
Storage	About five hours of peak output	Sold as a capacity product, separate from energy
Power cycle	Commercial supercritical steam	Shops and operators exist; sCO2 held as a later uprate
Net output	350 MWe firm, 500 MWe for 5 h	Firm energy plus a capacity product
Design life	60 years, with an 80-year watch item	Matches current U.S. license practice and coal-site wires
Capacity factor	92% reactor target	Storage, not the core, provides flexibility
Staff	200–250 FOAK, security included	Chemistry and security dominate; do not promise a 30-person plant

Figure. Forge-350 program-level planning positions. These are business targets, not design parameters; reactor values are set by the licensed vendor design.

7.1 Nuclear island

The nuclear island is bought, not invented. The owner selects a vendor design with a public NRC licensing record and writes program requirements, not hardware. Three items are long-lead for any salt-cooled pebble design and must be on the schedule from the first layout, not added after the cavity is poured. Online fuel handling lets the reactor stay at power, but it is also a safeguards and maintenance program. Graphite and high-temperature structural materials need ASME qualification, with any code case opened during pre-application, not after the construction permit; that qualification is a long-lead item equal to the vessel. Beryllium in the coolant is an industrial-hygiene constraint: the salt system is managed under a program written to OSHA and NRC expectations together.

7.2 Fuel

Forge-350 assumes TRISO pebble fuel of the kind Kairos and X-energy have described publicly. The fuel design belongs to the vendor and the fabricator, not to this guide. Fuel assay is a vendor and licensing decision fixed in pre-application. The program interest is narrower: that choice drives HALEU demand, the plant's NRC security category, and fuel cost, so the owner should not let the plant be locked to the top of the HALEU band.

The first core should be contracted as a DOE-allocated HALEU campaign plus a commercial backfill, with TRISO-X or a second qualified shop as fabricator. A plant that reaches construction with only a letter of intent for fuel is not a plant; Kemmerer's schedule risk is substantially a fuel risk. Spent fuel goes to licensed on-site dry storage using approved

storage and transport packages. Reprocessing is out of scope for unit 1. A later recycle decision is a separate national-policy question.

7.3 Power conversion, storage, and the coal site

The intermediate nitrate loop is the economic device. It keeps the primary salt at the heat exchanger, feeds a steam generator, and charges tanks the concentrating-solar industry already knows how to build. Crescent Dunes is the caution: salt freezes if heat tracing and operating procedures are treated as balance-of-plant trivia. Freeze protection and backup heat are safety-adjacent systems even if they are not safety class.

Siting is not a cosmetic choice. A retiring coal unit brings a switchyard, an interconnection, rail, a cooling-water right or a cooling-tower pad, a workforce, and often an energy-community tax position. The ADVANCE Act points the NRC at streamlined treatment of brownfield and coal sites; it does not waive seismology, hydrology, or emergency planning. Prefer a site with a completed interconnection study, a willing owner, and a state commission that has already accepted nuclear in its resource plan. The Tennessee Valley, the Ohio Valley, and inland Southeast coal stations are the natural first fleet because the wires, the operators, and the nuclear service industry are already there. Dry or hybrid cooling should be the base case wherever the coal plant's once-through withdrawal will not be re-permitted.

8. Safety case, in the form a reviewer expects

Forge-350 should not ask the NRC to take "walk-away safe" as a finding. It should ask the staff to review a short list of claims, each supported by the vendor's analysis and methods the staff already has. The claims below are licensing themes, not design specifications.

- **Functional containment.** The safety case relies primarily on the fuel and the coolant to retain radioactive material, rather than on a large high-pressure containment building. This claim is already the Kairos licensing theory; Forge-350 should adopt it only to the extent public Hermes data and NRC-reviewed fuel methods support it.
- **No large-break LOCA.** A low-pressure primary with a large margin to coolant boiling removes the accident that sizes light-water emergency core cooling. The licensing basis instead turns on a different set of events that the vendor must analyze and the staff must review.
- **Passive decay heat.** Decay heat removal should be passive, with no AC power and no operator action required for an extended period, and analysis carried to a stable end state. This matches what U.S. reviewers already accept on AP1000 and ESBWR, translated to salt.
- **Reactivity.** Inherent safety claims are cases to analyze, not features to assume. Reactivity control and shutdown are vendor design and NRC review items; the owner's requirement is that they be demonstrated by analysis, not by brochure.
- **Source term and emergency planning.** The design target is a plume emergency planning zone at the site boundary, justified by a mechanistic source term. The project budget and the site screen must still close if the NRC requires a conventional 10-mile zone. Do not select a site that only works under the optimistic zone.
- **External hazards.** Seismic and flood design bases are site-specific and set in the licensing analysis; screen sites conservatively, with flood margins above the coal plant's historical basis plus a climate allowance, and plan aircraft and vehicle barriers under 10 CFR 73. HALEU in the quantities a commercial core needs falls in a higher NRC security category than light-water fuel. Physical security is a cost line, not a brochure claim.

Beryllium toxicity and tritium are the two chemical issues that will draw intervenor attention. Both are manageable and neither is optional to analyze. Tritium control is a vendor design and licensing item the application must address explicitly. A beryllium hygiene program is an operating license exhibit.

9. Economics that can reach a credit committee

Capital is the product. On a light-water plant, fuel and operations are small beside the recovery of the build. Forge-350 does not escape that. It tries to spend the capital on fewer unique civil works, a low-pressure reactor that does not need a thick pressure vessel, and a site that does not need a new switchyard.

Case	Overnight capital	What it assumes
U.S. AP1000 actual (Vogtle 3 and 4)	$14,000–16,000/kW all-in	Completed 2023–24; ~$31–37 billion for 2,234 MW
NuScale CFPP at cancellation, 2023	About $20,000/kW	462 MW estimate near $9.3 billion; project stopped
Natrium public target, Kemmerer	About $11,000–12,000/kW	Up to ~$4 billion for 345 MWe, about half DOE cost share
BWRX-300 first unit, public TVA-range	About $18,000/kW	Figures near $5.4 billion before interest; later units hoped near $12,000/kW
DOE nth-of-a-kind ambition	About $3,600/kW overnight	After 10–20 units; not a first-plant forecast
Forge-350 FOAK target	$3.8–5.0 billion overnight	$11,000–14,000/kW on 350 MWe firm; storage included
Forge-350 NOAK target, unit 8+	$2.0–2.6 billion overnight	$6,000–7,500/kW on firm output, design frozen, coal site

Figure. Capital anchors in public 2023–2026 figures, and Forge-350 planning bands. Overnight cost excludes interest during construction.

Interest is the second price. Five years of construction at 8 percent can add roughly a fifth to a quarter on top of overnight cost. That is why the schedule in Section 10 is a financial control, not a public-affairs date. A first unit at $4.4 billion overnight plus financing, running at 92 percent, needs on the order of $90–140 per megawatt-hour to cover capital alone before fuel, staff, and decommissioning, depending on the cost of money and the recovery period. Fuel and operations might add $15–25. Unassisted, that is not a merchant energy-only price. Assisted, it can be a contracted price.

The revenue stack that makes unit 1 bankable in the United States is specific:

- **A long offtake** from a utility, a federal power marketer, or a large load (data center or industrial host) covering at least 70 percent of firm output for 20 years or more. Merchant exposure on unit 1 is how projects die in the credit committee.
- **A capacity payment** on the storage-backed 150 MWe, sold separately from energy. This is the Natrium commercial idea, and it is worth more in a market that pays for accredited capacity than in an energy-only market.
- **DOE cost share or a Title 17 loan guarantee** on the first unit only. Kemmerer's structure — public cost share on the order of half the FOAK premium — is the template. Later units should not need it, and the guide treats dependence on cost share past unit 2 as a kill signal.
- **Tax position.** New nuclear can qualify for the technology-neutral clean electricity production credit under section 45Y, or the investment credit under 48E, but not both on the same facility. With prevailing-wage and apprenticeship rules met, the production credit is on the order of 3 cents per kilowatt-hour after recent inflation adjustment (IRS notices for 2026 put the relevant alternative amount at 3.1 cents). A coal-closure energy community and domestic-content compliance can each add 10 percent. Full value for non-wind, non-solar technologies currently runs to facilities placed in service by the end of 2033, with a partial window into 2035. Section 45U is for existing reactors placed in service before the Inflation Reduction Act, not for Forge-350. Confirm the window with tax counsel. A Chinese state-owned owner, or a nuclear island that fails the prohibited-foreign-entity material-assistance test, disallows 45Y and 48E; the credit case here assumes allied or domestic content.

At those supports, a plausible contracted price for unit 1 is $70–110 per megawatt-hour in levelized terms, falling toward $50–70 for a later unit if the NOAK band is hit. Those are planning ranges, not forecasts. The reader should notice that they overlap the real cost of firm gas plus a carbon or reliability constraint, and that they do not overlap unsubsidized energy-only solar.

10. Implementation guide

The sequence below is a project-management and licensing sequence, written for an owner who does not yet have a site. Elapsed time from a serious start to commercial operation is eight to ten years if fuel and the test reactor cooperate, and longer if they do not. The 2026 permit reviews — about fourteen months for TVA's BWRX-300 application, and under two years from docketing to permit for Natrium — are the new benchmark. They are not a promise.

Gate	When	Exit criteria — do not pass without these
0 Owner	Month 0–6	Taxable owner, state partner, and a balance sheet that can carry $50–100 million of development cost. Joint venture acceptable; a shell company is not.
1 Site and offtake	Month 3–18	Two coal or brownfield sites under option. Interconnection position. Offtake term sheet for 70% of firm output. Water and transmission fatal-flaw screen.
2 Fuel	Month 6–18, before CP file	HALEU allocation or commercial enrichment contract, and a fabrication slot. If neither can be dated, switch the nuclear island to the light-water off-ramp.
3 Pre-application	Month 6–24	NRC engagement plan; white papers on functional containment, salt spill, fuel handling, and EPZ. ASME code-case plan. Owner's engineer appointed.
4 Construction permit	File ~month 24; review 14–24 months	PSAR, environmental report, safeguards. Limited work authorization for non-nuclear work. Hearing strategy budgeted.
5 FID and build	Month 42–90	Overnight cost band reconfirmed. EPC is multi-contract or has a replaceable nuclear constructor. Non-nuclear complete. Fuel delivery date inside the outage window.
6 Operating license	In parallel with build	FSAR, technical specifications, emergency plan, security plan, beryllium and tritium programs. ITAAC-equivalent closure.
7 Commercial	Month 96–120	Power ascension, warranty run, offtake conversion. NOAK design freeze within 12 months of commercial operation, or the fleet does not get cheaper.

Figure. Decision gates. A missed fuel gate switches technology. It does not slip the site.

10.1 Contracting

Vogtle concentrated risk in a consortium that could not absorb it. Kemmerer's 2026 EPC split showed the same single-point exposure on an advanced reactor. Forge-350 should be bid as packages: nuclear island, salt systems, turbine island, civil, and integration. The owner keeps an owner's engineer with stop-work authority. Liquidated damages are worth less than the right to retender a package. Fixed-price on a finished design is reasonable for units 3 and after. Fixed-price on unit 1 is how contractors put contingency in the number or walk away when the number is wrong.

10.2 Light-water off-ramp

At Gate 2 and again at Gate 4, the owner scores HALEU delivery, TRISO qualification, and Flibe supply as red, amber, or green. Two reds switch the nuclear island to a licensed light-water SMR on the same site, same offtake, and same interconnection — the BWRX-300 being the design with a U.S. construction permit and a Canadian lead unit. The storage tanks can still be installed and charged electrically or from a later salt unit. The off-ramp is a feature. A project that cannot abandon its favorite coolant is a project that will wait past the tax window.

10.3 Fleet, not a single plaque

Unit 1 is a demonstration that happens to sell power. Units 3 through 8 are the product. The implementation plan should name eight coal sites in two reliability regions before unit 1 pours nuclear concrete, even if only two are under option. Design changes after unit 1 commercial operation require a written reason. Every "improvement" that breaks serial fabrication is presumed rejected. This is the Korean and Chinese lesson that survives translation: the second unit is cheaper only if it is the same unit.

11. Regulatory path

File unit 1 under 10 CFR Part 50: construction permit, then operating license. That is the path Natrium and the Clinch River BWRX-300 actually used. Part 52 design certification is the right step for unit 3, once the as-built design has stopped moving. Part 53, when it is final and the staff has reviewed one application under it, can become the fleet path. Leading with an unfinished rule is how a project becomes the staff's learning case.

- Pre-application: topical reports on TRISO in salt (building on the vendor's NRC-reviewed fuel methods), salt chemistry, confinement versus containment, and a proposed principal design criteria set aligned with the NRC's advanced-reactor design criteria.

• Environmental report: coal-site reuse is the mitigation story — no new transmission corridor, reuse of intake where legal, remediation overlap with coal ash only if the plume interaction is actually analyzed. NEPA delay is a schedule risk equal to the safety review.

• Hearings: the ADVANCE Act and later bills have moved some uncontested hearing practice. Contested hearings have not been abolished. Budget 12 to 18 months of hearing time in the permit float rather than discovering it at the end.

• State: a certificate of public convenience and necessity or equivalent, water permit, air permit for auxiliary boilers and cooling towers, and a state emergency-planning agreement. A federal construction permit without a state certificate is not a start of construction.

• Security and material: the physical-protection category NRC rules assign to the HALEU quantities on site; transport packages for fresh and spent fuel licensed before the first shipment, not before the first outage.

Price-Anderson indemnity applies to a licensed plant and should be confirmed in the ownership agreement so a minority investor is not surprised by the channeling of liability. Decommissioning funding starts with a site-specific estimate at operating-license stage and a trust funded under the NRC's formula. Salt inventory disposition is part of that estimate; it is not free.

12. Supply chain that must exist before concrete

A reader who remembers only one list should remember this one. Each line has killed or slipped a real project.

• **HALEU supply.** Pilot kilograms at Piketon do not fuel a commercial core. Contract tonnes, with a second source or a DOE backstop, before the construction-permit filing.

• **TRISO fabrication.** One Part 70 license exists. A second shop, or a qualified second line in the first shop, is the difference between a fleet and a prototype.

• **Nuclear graphite.** Qualify the actual supplier. Substitution after the permit is a license amendment.

• **Coolant salt.** Reactor-grade salt is not a catalog chemical, and its specialty constituents have few qualified suppliers. Lead time is measured in years. Beryllium handling is a siting constraint for the supplier as well as the plant.

• **High-temperature code materials.** Open the ASME cases during pre-application. An unqualified alloy is a redesign.

• **Nitrate-salt tanks and heat tracing.** Buy the solar-thermal lessons, including the failures. Freeze-up is a plant trip, and a dirty freeze-up is a multi-month outage.

• **Turbine and generator.** A 350–500 MWe supercritical set is a commercial machine. Order it when the offtake is signed; it is no longer the long pole if the salt systems are real.

• **Operators.** A coal workforce transfers maintenance culture, not licensed-operator qualifications. The senior reactor operator pipeline starts at Gate 3, not at fuel load.

13. Risk register and kill criteria

Risk	What it does to the project	Kill or mitigate
HALEU not available in tonnes by Gate 4	Moves commercial operation past the 45Y window	Kill the salt island; take the light-water off-ramp
TRISO-in-salt data do not support the source term	Forces a conventional containment and a new cost	Redesign only if overnight cost stays inside the FOAK band; else off-ramp
Single EPC dispute	2026 Kemmerer lesson; months vanish in rebidding	Multi-package contracts; replaceable constructor
Interest during a slip past 6 years	Adds more cost than any efficiency gain removes	FID only with fuel date, permit, and offtake; no speculative start
Contested hearing on EPZ or beryllium	12–24 month delay	Site works under a 10-mile EPZ; hygiene program in the application
Salt freeze or tank leak	Long outage after commercial operation	Freeze protection and tank-leak recovery required in the bid
Cost share promised but not appropriated	FOAK capital hole	FID condition precedent: executed DOE agreement, not a press release
Chinese safety-class vessel, fuel, or crew used to hit the price	Collides with 50.38, Entity List, PFE credit rules, or access authorization	Kill that procurement. Allied forgings and a domestic crew, or stop

Figure. Risks that stop the project, as distinct from risks that merely need a procedure.

14. A twenty-four-month checklist for the reader

The guide is written so that a utility planner, a state energy office, or an investor can act without waiting for a vendor's brochure. The first two years are development, not construction.

- Name the owner. If the reader is a state office, pick the public-power or cooperative counterparty that can hold a license. The NRC licenses a person, not a concept.
- Screen retiring coal units for interconnection, land, flood, seismic category, and community position. Drop any site that only works with a reduced emergency planning zone.
- Ask the offtaker for a term sheet, not a memorandum. Data-center load is real only when the ramp and the curtailment terms are written down.
- Open a fuel dialogue with DOE's HALEU program and with the licensed fabricator in the same quarter. Track kilograms under contract, not announcements.
- Fund an owner's engineer to produce a site-specific overnight cost with a −20 / +40 percent FOAK band. Reject any estimate shown as a single number.
- Request an NRC pre-application meeting only after the site shortlist and the fuel path exist. A meeting with neither wastes the staff's calendar and the owner's credibility.
- Model 45Y and 48E as sensitivities, including a case with no credit. If the no-credit case cannot be carried by the offtake, the project needs cost share or it needs to wait for unit 3 of somebody else's design.
- Write the off-ramp into the partnership agreement so a technology switch does not require a unanimous consent that will never come.
- Engage the coal plant's operators and the nearest nuclear plant's training pipeline. The license will be won or lost on people as much as on salt.
- Publish the kill criteria. A project that cannot say what would stop it is asking its ratepayers for a blank check — the error Vogtle made legible.
- Reject any bid that reaches its price only with a Chinese state-owned designer, a Chinese safety-class vessel, Chinese fuel, or a Chinese nuclear crew. Price the allied-forging, domestic-crew case, or do not take the bid to the board.

15. Sources and limits

Operating-fleet capacity factor (91 percent in 2025) and fleet size (94 reactors, about 97 GW) are from EIA-based public summaries. Vogtle cost and schedule are from owner filings and contemporary reporting, which do not agree on a single dollar figure; this guide uses the $31–37 billion, $14,000–16,000 per kilowatt band. Natrium's construction permit (March 2026), 345 / 500 MWe rating, and DOE cost share are from NRC and DOE releases; the Bechtel EPC exit is from September 2026 trade reporting. TVA's Clinch River BWRX-300 construction permit is from the NRC action of 29 September 2026. Hermes licensing status and design approach are from NRC public records and the applicant's public descriptions. HALEU licensing status is from the NRC's public HALEU page. Chinese vendor cost claims, including about 17,000 yuan per kilowatt for Hualong One, are domestic vendor figures; the Sanmen and Haiyang comparison with Vogtle is the same-reactor evidence. Foreign-ownership limits are the NRC's April 2026 amendment to 10 CFR 50.38 implementing ADVANCE Act section 301; China is outside the OECD-and-India exception. CGN's Entity List status dates from 2019, and the 2016 criminal case is a matter of public Justice Department record. Prohibited-foreign-entity credit rules are the 2025 reconciliation statute and IRS Notice 2026-15. Other tax-credit structure is from 26 U.S.C. §§ 45U and 45Y and from 2026 IRS inflation notices.

What this guide does not do: it does not size a core, it does not calculate a source term, it does not specify fuel, components, or reactor parameters, it gives no design, fabrication, construction, or operating procedures, and it does not replace a vendor safety analysis. Forge-350 is the architecture the 2026 record supports — solid fuel, low pressure, coal-site wires, storage sold as capacity, a permit path the NRC has just shown it can finish, and a fuel gate that is allowed to say no. The plant gets built, if it gets built, by an owner who keeps those constraints tighter than the technology story.

16. Decision card

Use this card in a resource-planning or investment meeting. It is the guide compressed to the choices that change a go / no-go, not a substitute for the sections above.

Question	Answer this guide supports
What should unit 1 be?	One salt-cooled, TRISO-fueled module, 350 MWe firm, 500 MWe for five hours from nitrate-salt storage, on a retiring coal site.
Why not an AP1000?	Best safety maturity, worst recent U.S. cost. A second pair only makes sense on a frozen design with a warm supply chain — not as the efficiency or schedule fix.
Why not start with BWRX-300?	Start there if HALEU is red at Gate 2. Otherwise hold it as the off-ramp. It is licensable now and thermally inefficient.
Why not Natrium as specified?	Copy the storage split and the coal site. Do not copy sodium for a first fleet if a salt coolant can carry the same heat without a sodium-water reaction.
Why not buy the Chinese plant?	Hualong One and the Chinese AP1000 builds show what serial construction costs. A Chinese state owner cannot hold the license under 50.38, CGN is an Entity List and military-company problem, and Chinese safety-class content can zero out 45Y/48E. Copy the batch order and the frozen design. Do not import the vendor, the vessel, the fuel, or the crew.
What permit?	10 CFR 50 construction permit, then operating license. Part 52 after the design freezes. Part 53 only once the staff has used it.
What has to be signed before filing?	Offtake term sheet for at least 70 percent of firm output, a dated HALEU and TRISO path, and an owner’s engineer cost band — not a single number.
What kills the salt plant?	No tonnes of HALEU by Gate 4, a source term that forces a new containment outside the cost band, or cost share that exists only as a press release.
What does success look like?	Unit 1 commercial in the mid-2030s inside the FOAK band, and a frozen design copied onto named coal sites without a new coolant.

Bottom line for the reader. Do not wait for a perfect reactor, and do not import a Chinese one. File a low-pressure, TRISO-fueled module on a coal site, with storage on the intermediate loop, under a Part 50 construction permit, allied or domestic safety-class equipment, and a signed HALEU path. Copy China’s batch order and frozen design. If the fuel path fails, build the light-water plant the same site was screened for.

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